

Dimensionless Numbers

The ratios that govern microfluidic flow — a Semitree cheat sheet

At the microscale, a handful of dimensionless numbers tell you which forces win and how a device behaves. Each is a ratio of two competing effects.

Reynolds number

Re

$$Re = \rho \cdot v \cdot Dh / \mu$$

inertial vs. viscous — laminar when $Re \leq 2000$

Péclet number

Pe

$$Pe = v \cdot L / D$$

advection vs. diffusion — high Pe: flow outpaces mixing

Capillary number

Ca

$$Ca = \mu \cdot v / \gamma$$

viscous vs. interfacial — sets the droplet regime

Weber number

We

$$We = \rho \cdot v^2 \cdot L / \gamma$$

inertial vs. interfacial — droplet breakup / jetting

Bond number

Bo

$$Bo = \Delta\rho \cdot g \cdot L^2 / \gamma$$

gravity vs. interfacial — tiny at the microscale

Capillary length

Lc

$$Lc = \sqrt{\gamma / (\Delta\rho \cdot g)}$$

where gravity $\approx$ surface tension (~ 2.7 mm for water/air)

Variables

ρ density (kg/m³)

v mean velocity (m/s)

Dh hydraulic diameter (m)

μ dynamic viscosity (Pa·s)

L characteristic length (m)

D diffusion coeff. (m²/s)

γ surface tension (N/m)

$\Delta\rho$ density difference (kg/m³)

g gravity (9.81 m/s²)

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Standard, published equations. Always verify against primary sources for critical work.